Fiscal Year:	FY 2007	**Task Last Updated:**	FY 12/20/2007
PI Name:	Thomas, Aaron Ph.D.		
Project Title:	Transport Processes in Novel Oscillating Flow Systems for Separation and Containment Removal		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Dynamics and stability		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	amthomas@uidaho.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	208-882-9038
Organization Name:	University of Idaho		
PI Address 1:	Department of Chemical Engineering		
PI Address 2:	PO Box 441021		
PI Web Page:			
City:	Moscow	**State:**	ID
Zip Code:	83844-1021	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	12/01/2003	**End Date:**	09/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA GRC
Contact Monitor:	Balasubramaniam, R.	**Contact Phone:**	216-433-2878
Contact Email:	RBalasubramaniam@grc.nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007 per PI (6/07) NOTE: Received NCE to 6/30/2007 per PI (10/06)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NNC04GA77G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The use of oscillating flows as a novel separations process addresses the requirement of maintaining an acceptable atmosphere in habitable environments as designated by NASA's critical path roadmap. This study uses oscillating flows as a means to augment current separation methods to remove carbon dioxide, organic contaminants, and fine sub-micron particles from the atmosphere. Excessive airborne pollutants are a result of materials off-gassing, human metabolism, leaks, fires, and chemical spills and present a risk to the health and safety of the crew in a closed system. Properly controlling these species to below the spacecraft maximum allowable concentrations (SMAC) is a priority for long duration manned space missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Oscillating flows as a mechanical separation mechanism can be applied to any mixture that required the removal or purification of any species that has a different molecular diffusion coefficient than the rest of the mixture. The method is applicable to carbon dioxide removal from air as studied in this work or to hydrogen purification for fuel cells. Although oscillating flows may not totally replace existing separation mechanisms, it can certainly serve to remove the bulk of the impurities or pre-concentrate species for further treatment. This work is still in the proof-of-concept stage, but significant advances have been made in order to begin construction of a commercial device.
Task Progress:	Oscillating flows has been analyzed as a novel separation technique for removal or purification of gas species. For contaminant removal from the air, the proposed separation mechanism is mechanical and does not need to be replaced or regenerated as membranes and/or zeolites would eventually require. Previous studies have shown that a limited separation is possible for various gas species in tubes with are straight. Although oscillating flows in these straight tubes are able to provide a high throughput (mass transfer), the amount of separation achieved needed to increase to make the process a viable separation mechanism. This research focused on utilizing wavy tubes which bulge and constrict in a regular manner as a means to increase the separation of gas species. Altering the surface of the tubes will in turn alter the flow characteristics of the fluid. The shape of the flow was solved using analytical models that relied on a perturbation scheme, an approximation method. Results of the models showed that the wavy tubes produced flow recirculations in the bulges. The shape and magnitude of these recirculations depends on the frequency and amplitude of the flow oscillations, the relative radius of the tube, the viscosity of the fluid, and the size of the bulges and constrictions. These are the governing parameters that shape the flow profiles. The relative movement of dilute species present in oscillating flow was also analyzed using a perturbation scheme. It was shown that the throughput of species in wavy tubes is vastly different than the throughput in straight tubes. If two dilute species are present in an oscillating system, the separation between these two species is much greater for a wavy tube than for a straight tube. Indeed, the models show that the bulges and constrictions will lead to a higher separation between species. Experiments were then performed to verify the analytical findings. Although the experiments showed a higher throughput than predicted by the models, the amount of separation achieved was nearly identical to the theoretical predictions for several changes in the governing parameters. Because this work looked to separate CO2 and other contaminants from the air, a final experiment was done for 10% CO2 in air. The result was a 23 to 1 selectivity of oxygen to carbon dioxide using the oscillations in wavy tubes. There existed a slight separation of the nitrogen and oxygen in the air due to their different molecular diffusion. This shows that a high level of CO2 removal is possible with this system, and is a promising outcome to the success of human space exploration. A proposed mechanism to use flow oscillations on a long-manned space mission would be to connect two large reservoirs with a bank of wavy walled tubes. One reservoir would have an inlet of contaminated air, the other would have water vapor. The flow oscillations would be produced using either an oscillating piston or a pulsating bellow system. Because oxygen will move at a rate that is 23 times higher than carbon dioxide, the reservoir with the steam will be filled with the purified air. The steam can then be condensed out and regenerated air can then be recycled to the human systems or further treated if necessary. The other reservoir will then have a higher concentration of CO2 with some air and steam present as well. This can be treated or vented if desired. If further treatment is needed, this initial separation from the flow oscillations will greatly reduce the load on current contaminant removal methods. This will reduce the amount of resupply and regeneration required with current air revitalization techniques. As flows are continuously fed into and out of each reservoir, this process will be continuous that can handle large volumes and large flow rates if needed. It can also be tuned for the event other contaminants resulting from a chemical spill or fire need to be removed from the air as well. Although this is still in its beginning phases, the blueprint has been laid out to construct a viable apparatus for a given application. Work was also performed on using flow oscillations to enhance the transport of heat from a heat source to a heat sink using fluid contained connectors. Getting the heat from the source to the heat sinks could be done via heat pipes and metal conductors, but the mass of such equipment is high. Oscillating flows involves the use of connecting pipes that contain a fluid which is in perpetual motion, so that the heat can be conveyed fast. Models and experiments were researched on two different configurations, and open tube and an annulus, to analyze the best way for heat to be removed. Depending on what parameters were held fixed, different conclusions could be reached as to which provided the best heat removal. However, the flow oscillations and the two configurations showed a high rate of heat removal. Although the heat transfer experiment that was conducted follows the theoretical model well, it has yet to be seen whether this technology would work on a larger scale, since the volume of outer space modules are several orders of magnitude greater than the volume of the experiment conducted in the laboratory.
Bibliography Type:	Description: (Last Updated: 01/08/2008)
Abstracts for Journals and Proceedings	Oropeza I, Crain J, Kassab A, Divo E, Narayanan R. "Oscillatory Flow as a Means of Enhanced Species Separation – a Three Dimensional Time Accurate Numerical Study." Engineering and Applied Sciences, Porlamar-Margarita, Venezuela, March 2006. VIII International Congress on Numerical Methods in Engineering and Applied Sciences, March 2006. , Mar-2006

Abstracts for Journals and Proceedings	Thomas A, Thich G, Guthiller D, Narayanan R. "Separation of species by oscillatory flow in wavy-walled tubes." Habitation 2006 Conference, Orlando, FL, February 5-8, 2006. Habitation 2006;10(3-4):184-5. , Feb-2006
Abstracts for Journals and Proceedings	Volk J, Thomas A, Narayanan R, Kassab A, Divo E. "Heat removal from closed environments by a pulsating method." Habitation 2006 Conference, Orlando, FL, February 5-8, 2006. Habitation 2006;10(3-4):238. , Feb-2006
Abstracts for Journals and Proceedings	Divo E, Kassab A. "Air quality and ventilation in space vehicles and space environments." Habitation 2006 Conference, Orlando, FL, February 5-8, 2006. Habitation 2006;10(3-4):152-3. , Feb-2006
Abstracts for Journals and Proceedings	Thomas A, Thich G, Narayanan R. "Mass Transfer and Separation of Species in Oscillating Flows with a Wavy-Walled Boundary: An Analytical Study." APS Division of Fluid Dynamics, 58th Annual Meeting, Chicago, IL, November 2005. APS Division of Fluid Dynamics, 58th Annual Meeting, November 2005. , Nov-2005
Abstracts for Journals and Proceedings	Thomas A, Thich G, Narayanan R. "Mass Transfer and Separation of Species in Oscillating Flows with a Wavy-Walled Boundary: An Analytical Study." 2005 Annual Meeting, AIChE, Cincinnati, OH, November 2005. 2005 Annual Meeting, AIChE, November 2005. , Nov-2005
Abstracts for Journals and Proceedings	Thomas A, Narayanan R, Divo E, Kassab A, Mills T. "Use of Oscillatory Flow to Separate Species – A Method with Applications to Particle Separation." Obscurants and Smoke Workshop, Orlando, FL, June 2005. Obscurants and Smoke Workshop, June 2005. , Jun-2005
Articles in Peer-reviewed Journals	Thomas AM, Thich GK, Narayanan R. "Low Reynolds number flow in a channel with oscillating wavy-walls: An analytical study." Chemical Engineering Science, 2006 Sep;61(18):6047-56. http://dx.doi.org/10.1016/j.ces.2006.05.020 , Sep-2006
Dissertations and Theses	Volk, Johnathan. "Enhancing heat transport through oscillatory flows." Thesis, University of Florida, Gainesville, December 2006. , Dec-2006
Dissertations and Theses	Thich, Gatwech K. "Theoretical and Experimental Studies of Novel Gas Separation." Dissertation, University of Idaho, Moscow, December 2007. , Dec-2007
Dissertations and Theses	Guthmiller, Demian. "On the examination of the mass transfer and separation of dilute gases using oscillating flow with straight and wavy-walled tubes." M.S. Thesis, University of Idaho, 2005. , Dec-2005